

pdac-recurrent-cldn18-2-kras-g12r-tp53 -y220c-p25m

Selected general biomarker report — biomarker screening coverage and gaps

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Selected general biomarker report

Which biomarkers on the selected panel this patient has been tested for, and which have never been measured. The panel covers targets established in this tumor type, targets relevant across a subset of tumor types, and biomarkers that predict benefit regardless of where the tumor started. An unmeasured biomarker below is a gap in the record, not a prediction that the patient carries it.

Surveyed: 20 biomarkers. **No usable result on file:** 16. **Measured but not decision-grade:** 2. **Already established:** 2.

Twenty biomarkers were surveyed: sixteen never measured, and the two measured short of decision grade are the essential ones. Germline status is effectively unknown. The "none detected" on file is incidental tumor-NGS reporting, not the germline panel NCCN has recommended for every PDAC patient since 2019, and onset in the 50s sharpens it. BRCA1/2, PALB2, and ATM gate olaparib maintenance and platinum re-challenge. One blood draw, no tissue, an approved drug class behind a positive: the survey's highest-value order. The other essential gap, CLDN18.2, is a threshold call, not a yes-or-no. The tumor stained 2+/3+ in 60% of cells: below the 75% cutoff of zolbetuximab's registrational trials, clear of satri-cel's 2+ in 40%, ASP2138's 1+, and the radioligand protocol's 2+ in 20%, which 60% clears threefold. Which threshold applies decides which route is open; the clone went unnamed and the stain is archival.

Archival is the deeper problem: everything on file describes the December 2025 resection, not the July 2026 recurrence, with ten cycles of FOLFIRINOX in between. Two orders close the gaps: the blood draw (HLA class I typing rides along) and a recurrence biopsy, rank 1 for target validator and PI alike; one pass yields DNA and RNA sequencing and a six-stain block cut, CLDN18.2 re-stain included. Primary-to-recurrence concordance is 83.3%, but losses cluster in local recurrence and liver metastasis, this patient's setting, and biopsy sensitivity is 54.6%, so a negative stain on a small core is weak evidence of true loss.

The two finished results are answers, not gaps: microsatellite stability at TMB 0.83 mut/Mb closes single-agent PD-1 blockade, and the ranking says so. KRAS G12R is confirmed, but no G12C- or G12D-selective agent covers it. KMT2C S2816fs stays at the user's request despite no clinical route, and everything else on the panel is absent on purpose. Most of what remains will test negative. The case for testing is the size of the option a positive opens, not an implied likelihood of one.

Gaps: biomarkers with no usable result on file

HER2 protein overexpression (IHC 3+)

Priority: high **Scope:** Tumor-agnostic **Status:** Not measured

HER2 IHC 3+ carries a tumor-agnostic approval, so the primary site does not enter into it. In pancreatic adenocarcinoma the rate is low, on the order of a few percent, and HER2 has never been stained here.

HER2 has never been stained in this case, and an IHC 3+ result would open an approved tumor-agnostic ADC in a patient whose disease recurred four months after adjuvant chemotherapy. The honest expectation is a negative result, since only a small percentage of pancreatic adenocarcinomas are HER2 IHC 3+, but the stain is inexpensive, runs on tissue that already exists, and needs no new procedure. A 2+ result reflexes to ISH and does

not qualify for the tumor-agnostic indication on its own.

Recommended assay. HER2 IHC with reflex ISH on a 2+ result (IHC · archival December 2025 resection block, or recurrent-lesion tissue if the recurrence is biopsied · under 1 week once tissue is in hand)

Bundles with. single IHC block cut (CLDN18.2, HER2, mesothelin, TROP2, B7-H3, CEACAM5) on recurrent-lesion tissue if biopsied, otherwise on the December 2025 resection block

A positive result would open.

- trastuzumab deruxtecan (HER2-directed ADC · tumor-agnostic approval · NCT04482309)

Next step. Bundle with planned testing

References. nct:NCT04482309;
fda:fam-trastuzumab-deruxtecan-nxki-HER2-IHC-3+-solid-tumors-accelerated-approval-2024

Mesothelin (MSLN) protein expression

Priority: high **Scope:** Relevant to this tumor type **Status:** Not measured

Mesothelin is expressed in roughly 80 to 90 percent of pancreatic ductal adenocarcinomas against very limited normal expression, which makes it the unmeasured surface antigen most likely to be positive in this patient.

Mesothelin has not been stained, and it is the surface target most likely to be positive in this tumor type. A positive stain makes the mesothelin CAR-T, ADC, immunotoxin, and bispecific programs screenable, none of which is approved and several of which take pancreatic adenocarcinoma specifically. It costs one more stain on tissue that already exists, so the gap is cheap to close and awkward to leave open if cell-therapy trials are being considered.

Recommended assay. Mesothelin IHC on tumor tissue (IHC · archival December 2025 resection block, or recurrent-lesion tissue if biopsied · under 1 week once tissue is in hand)

Bundles with. single IHC block cut (CLDN18.2, HER2, mesothelin, TROP2, B7-H3, CEACAM5) on recurrent-lesion tissue if biopsied, otherwise on the December 2025 resection block

A positive result would open.

- anetumab ravtansine (mesothelin-directed ADC · trial only)

Next step. Bundle with planned testing

References. pmid:32605175

NTRK1 / NTRK2 / NTRK3 gene fusion

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

Three TRK inhibitors are approved regardless of primary site, so the only question is whether this tumor was ever run on an assay capable of finding a fusion.

No fusion result is on file, and whether the December 2025 panel covered fusions is not stated, so retrieving that

report is the cheapest first move. TRK fusions in pancreatic adenocarcinoma are rare and are largely confined to KRAS wild-type tumors, and this tumor carries KRAS G12R, which makes a positive result very unlikely. It reads out at no extra cost on the RNA-capable panel already recommended for re-profiling the recurrence, so it does not justify a separate order.

Recommended assay. RNA-based fusion NGS covering NTRK1/2/3 (NGS_panel · recurrent-lesion biopsy tissue; archival tissue or ctDNA are fallbacks with lower fusion sensitivity · 2-3 weeks)

Bundles with. recurrent-lesion biopsy: one pass supplies comprehensive DNA plus RNA tumor NGS and a block for the IHC stains below

A positive result would open.

- larotrectinib (TRK inhibitor · tumor-agnostic approval · NCT02576431)
- entrectinib (TRK / ROS1 / ALK inhibitor · tumor-agnostic approval · NCT02568267)

Next step. Bundle with planned testing

References. nct:NCT02576431; nct:NCT02568267

RET gene fusion

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

RET fusion-positive solid tumors have a tumor-agnostic approved option irrespective of primary site, and no RET result appears anywhere in this record.

As with NTRK, no RET result exists and RET fusions in pancreatic adenocarcinoma occur in KRAS wild-type tumors, which this is not. What changed recently is the strength of the option behind a positive result: selpercatinib converted from accelerated to traditional approval in July 2026, so it is now a fully approved tumor-agnostic drug rather than a provisional one. The test rides on the same RNA-capable panel and costs nothing extra.

Recommended assay. RNA-based fusion NGS covering RET (NGS_panel · recurrent-lesion biopsy tissue; ctDNA is a lower-sensitivity fallback for fusions · 2-3 weeks)

Bundles with. recurrent-lesion biopsy: one pass supplies comprehensive DNA plus RNA tumor NGS and a block for the IHC stains below; same assay as the NTRK row

A positive result would open.

- selpercatinib (selective RET inhibitor · tumor-agnostic approval · NCT03157128)

Next step. Bundle with planned testing

References. nct:NCT03157128; fda:selpercatinib-RET-fusion-positive-solid-tumors-traditional-approval-2026-07

B7-H3 (CD276) protein expression

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

B7-H3 is over-expressed across many solid tumors including pancreatic adenocarcinoma, with relatively low normal-tissue surface expression. It is a pan-cancer target that a panel ordered to answer a pancreatic question would not report.

B7-H3 has not been stained. Expression would make the B7-H3 ADC and CAR-T programs screenable, and several of those take solid tumors broadly rather than by site. Nothing behind this target is approved anywhere, so the row buys trial access rather than treatment, but it rides on the same block cut as the other stains.

Recommended assay. B7-H3 (CD276) IHC on tumor tissue (IHC · archival December 2025 resection block, or recurrent-lesion tissue if biopsied · 1-2 weeks once tissue is in hand)

Bundles with. single IHC block cut (CLDN18.2, HER2, mesothelin, TROP2, B7-H3, CEACAM5) on recurrent-lesion tissue if biopsied, otherwise on the December 2025 resection block

A positive result would open.

- ifinatamab deruxtecan (B7-H3-directed ADC · trial only)
- enoblituzumab (anti-B7-H3 mAb · trial only)

Next step. Bundle with planned testing

References. pmid:34349762

Fibroblast activation protein (FAP) expression on cancer-associated fibroblasts

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

Pancreatic adenocarcinoma is the classic desmoplastic tumor and FAP marks its activated cancer-associated fibroblasts rather than the malignant cells. FAPI PET reads it out with no tissue at all, which is worth noting when the recurrence so far exists only on imaging.

FAP is unmeasured, though in this tumor type stromal expression is closer to expected than uncertain, so a result would carry less information than the tumor-cell targets on this list. FAP radioligand and bispecific programs are investigational and FAPI PET is available mainly through research protocols in the United States. Raise it only if a FAP-directed trial is in play, where its advantage is needing no biopsy.

Recommended assay. FAPI PET/CT, or FAP IHC on tumor tissue (imaging · none for FAPI PET; archival or recurrent-lesion tissue for IHC · depends on protocol access)

A positive result would open.

- ¹⁷⁷Lu-FAP-2286 (FAP-directed radioligand · trial only)

Next step. Defer

HLA class I genotype (4-digit)

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

None of the HLA-presented antigens in the selected panel is established in pancreatic adenocarcinoma, so the

value here is access rather than antigen biology: HLA-restricted TCR-T and ImmTAC products, including the TCR approaches this case's KRAS G12R feature points toward, screen on allele type before anything else is considered.

There is no HLA type on file, which leaves every HLA-restricted cell therapy unassessable rather than excluded. It is a blood test that can ride along with the germline panel draw, and one result either opens or closes a whole class of trials. Roughly half of patients of European ancestry carry HLA-A*02:01, the allele most of these products are built around, so the odds of the result being useful are not small.

Recommended assay. HLA class I genotyping at 4-digit (allele-level) resolution (hla_typing · whole blood or buccal swab; no tumor tissue · 1-2 weeks)

Bundles with. one blood draw shared with germline multigene panel and HLA class I typing

A positive result would open.

- HLA-restricted TCR-T and ImmTAC agents (TCR-T / bispecific TCR · trial only)

Next step. Bundle with planned testing

TROP2 (TACSTD2) protein expression

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

TROP2 is broadly over-expressed across epithelial tumors and two TROP2 ADCs are approved in other tumor types, so its status here is worth knowing even though pancreatic adenocarcinoma is on neither label.

TROP2 has not been stained here. The caveat that matters is that both approved TROP2 ADCs are given without an expression cutoff in their labelled indications, and neither label includes pancreatic cancer, so a positive stain does not create eligibility for an approved use. Its value is in supporting a trial screen or an off-label argument, which puts it below the CLDN18.2 and mesothelin stains in the same block cut.

Recommended assay. TROP2 IHC on tumor tissue (IHC · archival December 2025 resection block, or recurrent-lesion tissue if biopsied · 1-2 weeks once tissue is in hand)

Bundles with. single IHC block cut (CLDN18.2, HER2, mesothelin, TROP2, B7-H3, CEACAM5) on recurrent-lesion tissue if biopsied, otherwise on the December 2025 resection block

A positive result would open.

- sacituzumab govitecan-hziy (TROP2-directed ADC · approved, other tumor type)
- datopotamab deruxtecan (TROP2-directed ADC · approved, other tumor type)

Next step. Bundle with planned testing

References. pmid:38948057

CEACAM5 (CEA) protein expression

Priority: medium **Scope:** Relevant to this tumor type **Status:** Not measured

CEACAM5 is expressed in pancreatic adenocarcinoma alongside colorectal and gastric disease. Serum CEA is often followed in this setting while the tissue antigen itself goes unmeasured, and only the latter gates the binder programs.

Neither tissue CEACAM5 nor a serum CEA value appears in the record. Tissue expression is what the CEACAM5 ADC and T-cell-engager trials screen on; serum CEA is a separate and weaker measurement that tracks disease rather than eligibility, and it is worth having for its own reasons. The stain rides on the block cut, and the programs behind it are all investigational.

Recommended assay. CEACAM5 IHC on tumor tissue (IHC · archival December 2025 resection block, or recurrent-lesion tissue if biopsied · 1-2 weeks once tissue is in hand)

Bundles with. single IHC block cut (CLDN18.2, HER2, mesothelin, TROP2, B7-H3, CEACAM5) on recurrent-lesion tissue if biopsied, otherwise on the December 2025 resection block

A positive result would open.

- tusamitamab ravtansine (CEACAM5-directed ADC · trial only)
- cibisatamab (CEA x CD3 bispecific · trial only)

Next step. Bundle with planned testing

Cadherin-17 (CDH17) protein expression

Priority: medium **Scope:** Relevant to this tumor type **Status:** Not measured

CDH17 is a gastrointestinal-lineage antigen expressed in pancreatic as well as colorectal and gastric adenocarcinoma, with normal expression confined to the GI tract.

CDH17 is unmeasured. Its confinement to gastrointestinal lineage is what makes it attractive as a CAR-T antigen, but the programs are phase 1, so a positive stain buys a trial conversation rather than a treatment. It is not a routine stain in most laboratories, so add it to the block cut when a CDH17 protocol is actually under consideration.

Recommended assay. CDH17 IHC on tumor tissue (IHC · archival December 2025 resection block, or recurrent-lesion tissue if biopsied · 1-2 weeks; not stocked by every laboratory)

Bundles with. single IHC block cut (CLDN18.2, HER2, mesothelin, TROP2, B7-H3, CEACAM5) on recurrent-lesion tissue if biopsied, otherwise on the December 2025 resection block

A positive result would open.

- CDH17 CAR-T (CAR-T · trial only)

Next step. Order if tissue available

Integrin beta-6 (alpha-v beta-6) expression

Priority: medium **Scope:** Relevant to this tumor type **Status:** Not measured

Integrin alpha-v beta-6 is induced on pancreatic carcinoma cells and is largely absent from normal adult

epithelium, which is one of the better tumor-versus-normal contrasts available in this tumor type.

No alpha-v beta-6 result is on file. Two integrin beta-6 ADCs are in trials, one of them in phase 3 in other tumor types, and both select on expression. The stain is not routine outside specialist laboratories, so order it when a specific protocol asks for it rather than pre-emptively.

Recommended assay. Integrin beta-6 (alpha-v beta-6) IHC on tumor tissue (IHC · archival December 2025 resection block, or recurrent-lesion tissue if biopsied · 2-3 weeks; specialist or sponsor-provided assay)

Bundles with. single IHC block cut (CLDN18.2, HER2, mesothelin, TROP2, B7-H3, CEACAM5) on recurrent-lesion tissue if biopsied, otherwise on the December 2025 resection block

A positive result would open.

- sigvotatug vedotin (integrin beta-6-directed ADC · trial only)
- JSKN022 (integrin beta-6-directed ADC · trial only)

Next step. Order if tissue available

Prostate stem cell antigen (PSCA) expression

Priority: medium **Scope:** Relevant to this tumor type **Status:** Not measured

PSCA is up-regulated in pancreatic ductal adenocarcinoma as well as prostate and gastric cancer, and PSCA cell-therapy programs have enrolled pancreatic adenocarcinoma specifically rather than by extrapolation.

PSCA is unmeasured. Cell therapy against it has been taken into pancreatic adenocarcinoma directly, which is more than can be said for most targets on this list, but the constructs are early phase and one lead program has stopped. Order the stain once a live PSCA protocol is identified rather than on spec.

Recommended assay. PSCA IHC on tumor tissue (IHC · archival December 2025 resection block, or recurrent-lesion tissue if biopsied · 1-2 weeks once tissue is in hand)

Bundles with. single IHC block cut (CLDN18.2, HER2, mesothelin, TROP2, B7-H3, CEACAM5) on recurrent-lesion tissue if biopsied, otherwise on the December 2025 resection block

A positive result would open.

- PSCA CAR-T (City of Hope) (CAR-T · trial only)
- BPX-601 (CAR-T · trial only)

Next step. Order if tissue available

References. pmid:42103354

BRAF V600E mutation

Priority: low **Scope:** Tumor-agnostic **Status:** Not measured

BRAF V600E is actionable across primary sites with combined BRAF and MEK inhibition, and no BRAF result

appears in this record either way.

No BRAF call is in the record, although a panel that reported KRAS codon 12 almost certainly covered BRAF codon 600, so pulling the December 2025 report is the cheapest way to close this. BRAF V600E and KRAS mutations are effectively mutually exclusive and this tumor carries KRAS G12R, so a positive result would be surprising. It reads out at no cost on any repeat sequencing.

Recommended assay. Tumor NGS covering BRAF codon 600 (NGS_panel · recurrent-lesion tissue or ctDNA; the original December 2025 report may already answer it · immediate if the prior report is retrieved; 2-3 weeks on new sequencing)

Bundles with. recurrent-lesion biopsy: one pass supplies comprehensive DNA plus RNA tumor NGS and a block for the IHC stains below

A positive result would open.

- dabrafenib plus trametinib (BRAF inhibitor plus MEK inhibitor · tumor-agnostic approval · NCT02034110)

Next step. Bundle with planned testing

References. nct:NCT02034110

PD-L1 protein expression

Priority: low **Scope:** Relevant across a tumor subset **Status:** Not measured

PD-L1 gates or enriches checkpoint benefit in several tumor types, but pancreatic adenocarcinoma is not among them: no PD-L1 cutoff governs an approval in this disease.

PD-L1 has not been stained, and in pancreatic adenocarcinoma no approval or established regimen turns on the result. With microsatellite-stable status and TMB at 0.83 mutations per megabase, the checkpoint routes PD-L1 would inform are already closed. Order it only when a specific trial names it as an eligibility criterion, in which case the protocol's clone and scoring system are what matter.

Recommended assay. PD-L1 IHC using the clone and scoring system named by the protocol requesting it (IHC · archival or recurrent-lesion tissue · under 1 week once tissue is in hand)

A positive result would open.

- PD-1 / PD-L1 inhibitors (checkpoint inhibitor · approved, other tumor type)

Next step. Defer

ROR1 protein expression

Priority: low **Scope:** Relevant across a tumor subset **Status:** Not measured

ROR1 is an onco-fetal receptor re-expressed in pancreatic carcinoma among other solid tumors, with essentially no adult normal expression.

ROR1 is unmeasured. The clinical programs have concentrated on B-cell malignancies and the solid-tumor CAR-T that would apply here has stopped enrolling, so the therapeutic tail in pancreatic adenocarcinoma is thin at present. Revisit if a ROR1 solid-tumor protocol opens.

Recommended assay. ROR1 IHC on tumor tissue (IHC · archival or recurrent-lesion tissue · 2-3 weeks; specialist assay)

Bundles with. single IHC block cut (CLDN18.2, HER2, mesothelin, TROP2, B7-H3, CEACAM5) on recurrent-lesion tissue if biopsied, otherwise on the December 2025 resection block

A positive result would open.

- zilovertamab vedotin (ROR1-directed ADC · trial only)
- LYL797 (ROR1 CAR-T · trial only)

Next step. Defer

MUC1 protein expression (including aberrant Tn-MUC1 glycoform)

Priority: low **Scope:** Relevant to this tumor type **Status:** Not measured

MUC1 is over-expressed and aberrantly glycosylated in more than 90 percent of pancreatic adenocarcinomas. The tumor-specific epitope is the glycoform rather than the protein, which changes what a test would have to measure.

MUC1 has not been assessed, and routine MUC1 IHC would not answer the question that matters, because the Tn-MUC1 cell therapies target an aberrant glycoform that a standard antibody does not separate from normal mucin. Near-universal expression also means a positive standard stain carries little information. Defer unless a MUC1 protocol names its own assay, which these programs generally do.

Recommended assay. Tn-MUC1 glycoform IHC (5E5-type antibody), not standard MUC1 IHC (IHC · archival or recurrent-lesion tissue · protocol-dependent; usually sponsor-provided)

A positive result would open.

- tecemotide (MUC1 peptide vaccine · trial only)

Next step. Defer

Measured, but not to decision resolution

A result exists for each of these, but not at the resolution a treatment decision needs. The workup that would close each gap is carried into the Target validation paths report.

Homologous recombination deficiency / BRCA1 / BRCA2 pathogenic variant (germline and somatic)

Priority: essential

On file. profile.json:biomarkers[7] 'Germline findings (incidental)' value 'none detected, per incidental germline reporting on the tumor NGS report', confirmation_status ngs_pending; profile.json:biomarkers[] reports KRAS, TP53, and KMT2C from the December 2025 tumor NGS; no BRCA1/2, PALB2, ATM, or genomic-instability score appears either way

Gap. Incidental germline reporting on a tumor NGS report is not a germline panel: it is not designed for germline sensitivity, it misses large rearrangements, and the report on file does not state which homologous-recombination genes were sequenced. Closing this takes a dedicated germline multigene panel on blood or saliva covering BRCA1, BRCA2, PALB2, ATM, CDKN2A, STK11, and the mismatch-repair genes, plus somatic HR gene coverage and a genomic-instability score on whatever tumor NGS is run next.

Germline status here rests on incidental reporting from a tumor NGS report, which is not a substitute for a germline panel, and guidelines call for germline testing in every patient with pancreatic adenocarcinoma. A pathogenic BRCA1/2 or PALB2 variant would put an approved PARP-inhibitor maintenance strategy on the table in this tumor type, argue for keeping platinum in the sequence, and change screening for relatives. Most patients test negative, and this is a blood draw with no biopsy attached, which is why it should go out now rather than wait on tissue.

CLDN18.2 protein expression (CLDN18 splice variant 2)

Priority: essential

On file. profile.json:biomarkers[0] CLDN18.2, '2+/3+ membranous staining in 60% of tumor cells (resected primary, December 2025 specimen)', confirmation_status confirmed; profile.json:biomarkers[1] 'CLDN18.2 (recurrent lesion)' not assessed, confirmation_status ihc_pending; profile.json:targetable_features[0]

Gap. Three things sit between this stain and an eligibility call. The result is 60% of tumor cells at 2+/3+, below the $\geq 75\%$ used in SPOTLIGHT and GLOW. The antibody clone and assay were not named, so it is not known whether this was the VENTANA CLDN18 (43-14A) RxDx assay that defines that cutoff. And the tissue is the December 2025 resection, not the recurrence that emerged after ten cycles of FOLFIRINOX. Re-staining with the CDx assay, ideally on recurrent-lesion tissue, closes all three at once.

The tumor is CLDN18.2-positive but at 60% of cells rather than the $\geq 75\%$ used in the registrational trials, on an unnamed assay, and on tissue that predates ten cycles of FOLFIRINOX. A 60% result is a weaker predictor of benefit than a $\geq 75\%$ result would be, and expression is patchy enough within a tumor that one archival block may not describe the recurrence. Re-staining, ideally on a recurrence biopsy, decides whether the CAR-T, ADC, and bispecific programs that set their own lower cutoffs are reachable, and it is the assay this case turns on.

Already established

- **Microsatellite instability / mismatch-repair deficiency (MSI-H / dMMR)** — profile.json:biomarkers[5] 'Microsatellite instability (MSI)' value 'negative (MSS)', method 'reported with tumor NGS panel (assay not specified)', confirmation_status confirmed; profile.json:targetable_features[4] carries MSS and TMB-low as a decision-relevant negative
- **Tumor mutational burden, high (TMB-H)** — profile.json:biomarkers[6] 'Tumor mutational burden (TMB)' value '0.83 mutations/Mb (low)', method 'tumor NGS panel (assay not specified)', confirmation_status confirmed; profile.json:targetable_features[4]

Panel targets out of scope for this tumor

61 of 72 panel targets were screened and set aside as having no plausible connection to this patient's tumor type or stated features: ALPPL2 · ANTXR1 · AXL · CA9 · CD19 · CD200 · CD37 · CD46 · CD70 · CDCP1 · CDH3 · CDH6 · CLEC12A · CT83 · CTAG1B · DLK1 · DLL3 · EGFR · EGFR (variant III) · ENPP3 · EPCAM · EPHA2 · EPHB4 · ERBB2 · ERBB3 · F3 · FOLH1 · FOLR1 · FZD7 · GPC3 · GPRC5D · GUCY2C · HAVCR1 · IL13RA2 · IL3RA · ITGA3 · L1CAM · LGR5 · LILRB4 · LY6E · LYPD3 · MAGEA1 · MAGEA4 · MAGEC2 · MERTK · MET · MMP14 · NECTIN4 · PODXL · PRAME · PTK7 · SEZ6 · SLC34A2 · SLC44A4 · SLITRK6 · SSTR2 · SSX2 · STEAP1 · TNFRSF10B · TNFRSF17 · TSPAN8